

# Behavioral Finance Research Project

Why Markets Aren't Rational, and What You Can Do About It

A project guide covering research design, methodology, five real-world case studies, the psychology behind each one, the milestone-by-milestone decisions made while building it, the real (not simulated) results obtained, and how the findings translate into practice.

Prepared as a companion guide to the code repository *research/behavioral-finance/*  
Written in English for readers without a quant-finance background; every concept is introduced with a plain-language example before it is used.

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## Executive Summary

Financial theory has long assumed that markets are, to a first approximation, rational: prices reflect available information, and systematic mispricing should be rare and quickly corrected by sophisticated investors. This project tests that assumption from two directions at once, using real code and, wherever technically possible, real data rather than narrative alone.

**Direction one (diagnosis):** why do sophisticated, model-driven institutions sometimes fail catastrophically, precisely because their models assumed rational, well-behaved markets? Long-Term Capital Management (1998), the "Quant Quake" (2007), Amaranth Advisors (2006), and the 2008 financial crisis are examined as cases where a locally reasonable model broke because it could not see how fast correlation and panic move together once other market participants stop behaving independently.

**Direction two (exploitation):** can well-documented behavioral biases — anchoring, underreaction, overreaction — be turned into a measurable, tradable edge, the way real firms such as Fuller & Thaler Asset Management and LSV Asset Management have built businesses on doing? This project builds a signal library and backtest engine, and runs it against a real 21-year daily price history for 48 India-listed companies (the only market data source technically reachable from the environment this project was built in — see Part V for why that constraint mattered and how it was worked around).

The honest headline result from that empirical run: a blended "behavioral score" combining three signals looked like a roughly break-even strategy — and that blend was hiding two very different stories. One component signal (anchoring to the 52-week high) lost money persistently and dramatically; another (short-term reversal) at first looked like a real, positive, cost-adjusted edge. The project's central methodological lesson — never trust a blended score without checking what is inside it — came directly out of this finding, not from a textbook. (Chasing *why* the losing component lost money turned into the project's longest, most instructive thread — Part V walks through five rounds of increasingly rigorous testing. The losing signal's cause turned out simpler, and more humbling, than any of the behavioral explanations first considered — and the same rigor, once finally pointed at the "winning" signal too, retracted it and surfaced a different, genuine finding instead.)

The guide that follows explains every concept used (behavioral and quantitative) in plain language with real examples, walks through all five case studies in detail, narrates the project's milestones and the reasoning behind each decision, and translates the findings into an investment framework, a risk-management playbook, and a business idea.

# Part I — Why This Project Exists

## 1.1 The theory markets are supposed to follow

The Efficient Market Hypothesis (EMH), the dominant framework in academic finance since the 1960s-70s, holds that asset prices reflect all available information at essentially every moment. Its practical implication is stark: if EMH holds strictly, no strategy should reliably beat a passive index over time, because any predictable pattern would immediately be traded away by rational, profit-seeking investors the moment it appeared.

EMH is not a strawman invented for this project to knock down — it is a genuinely useful default. Most of the time, for most large, liquid securities, prices really do incorporate public information quickly. The interesting question is not "is EMH ever true" (it often is, approximately) but "when and how does it fail, and can those failures be anticipated, measured, or profited from?"

## 1.2 Two stories that motivate the whole project

### TWO EPISODES THAT FRAME THE PROJECT

In 1998, a hedge fund whose partners included two Nobel-laureate economists (Myron Scholes and Robert Merton, co-creators of the option-pricing theory taught in every finance program) lost about 90% of its capital in roughly six weeks, despite running trades its own models considered close to riskless. In January 2021, a struggling video-game retailer's stock rose by roughly an order of magnitude in a few weeks, not because of any change in the company's prospects, but because a large, loosely coordinated group of retail traders on a Reddit forum decided to buy it — and forced some of the most sophisticated short-selling hedge funds in the world to take large losses covering their bets. Both episodes are covered in full in Part IV.

These two stories point in opposite directions — one is about a sophisticated model failing, the other about an unsophisticated crowd winning — but they share a single underlying fact: humans, not equations, decide what a price is. A model can be mathematically elegant and still miss the fact that, under stress, people (whether panicking institutional traders or excited retail traders) start behaving in a correlated, non-independent way that no amount of historical calibration prepared it for.

## 1.3 The project's two-track design

Two concrete, falsifiable questions structure everything that follows:

- **Q1 (diagnosis).** Can the mechanism by which rational-market risk models fail under stress be shown mechanically, with real numbers, rather than just asserted narratively?
- **Q2 (exploitation).** Do specific, named behavioral biases produce a real, measurable, cost-adjusted return premium when turned into a systematic trading signal, over a long sample, on real market data?

**DECISION & RATIONALE**

The project could have been written as a pure literature review — summarizing what Kahneman, Tversky, Thaler, and others found, and describing LTCM and GameStop narratively. That would have been faster, but it would not have produced anything falsifiable: a narrative account can always be told to confirm whatever conclusion the writer wanted. The decision here was to require runnable code and, wherever technically possible, real executed numbers for every claim — even when (as happened repeatedly, see Part V) the environment made that harder than expected. A claim in this project is either backed by a number that was actually computed, or it is explicitly flagged as not yet validated. Nothing in between.

## Part II — Core Behavioral Concepts, Explained Before They're Used

Every case study and every trading signal in this project rests on one of the concepts below. Each is introduced with an everyday example first, then the specific market mechanism it produces, so that later sections can use the term without re-explaining it.

### 2.1 Anchoring

**Everyday example:** if a car salesperson opens negotiation at \$30,000, most buyers end up settling closer to that number than if the same car had been opened at \$25,000 — even though the opening number carries no real information about the car's actual worth. The first number seen becomes a psychological reference point (an "anchor") that subsequent judgments are adjusted from, insufficiently.

**Market mechanism:** investors use a stock's 52-week high as exactly this kind of reference point. Even when good news arrives that should push the price well past the old high, many investors hesitate to buy through a level they've mentally anchored on as "expensive," so the price drifts up slowly instead of jumping — the mechanism behind the 52-week-high signal used in this project's Q2 signal library (Part III) and central to the George & Hwang (2004) academic finding it's based on.

### 2.2 Herding (Social Proof)

**Everyday example:** seeing a long line outside one restaurant and an empty one next door, most passersby join the line — not because they have independently evaluated either restaurant's food, but because a crowd is itself treated as evidence of quality. This is useful most of the time (crowds often do know something) and can be badly wrong when the crowd itself started for an unrelated reason.

**Market mechanism:** in a crisis, investors and institutions sell what everyone else is selling, and buy what's flowing toward safety (a "flight to quality"), regardless of the specific fundamental merits of what they're selling or buying. This is the central mechanism in the LTCM case (Part IV): LTCM's own trades were not herding, but the rest of the market's correlated flight to safety broke the correlation assumptions LTCM's model depended on.

### 2.3 Overconfidence and Illusion of Control

**Everyday example:** most drivers rate themselves as "above average" — a statistical impossibility for a majority to be true, and a well-replicated psychological finding. A string of good outcomes (safe drives, in this analogy) is easy to misread as evidence of skill rather than of favorable conditions that happened to hold.

**Market mechanism:** a trader with a strong track record under one market regime can mistake that track record for proof their model handles all regimes, and size positions accordingly. This is the central mechanism in the Amaranth Advisors case (Part IV): trader Brian Hunter's prior profitable natural-gas positions preceded a concentrated bet sized well beyond what the fund could survive being wrong about.

## 2.4 Underreaction and Overreaction

**Everyday example:** told a piece of surprising news, some people barely update their prior opinion at all (underreaction — "I'll believe it when I see more evidence"), while others swing to the opposite extreme immediately (overreaction — "this changes everything"). Both are documented, opposite failures of the same rational-updating standard (Bayesian updating: revising a belief by exactly as much as new evidence warrants, no more, no less).

**Market mechanism, underreaction:** after a company reports a surprising earnings beat, its stock price often keeps drifting upward for weeks afterward instead of jumping immediately to the new fair value — a well-documented anomaly called post-earnings-announcement drift (PEAD), because analysts and investors update their models too slowly. This underreaction is also the mechanism behind price *momentum* more generally: information diffuses slowly, so a trend already underway tends to continue.

**Market mechanism, overreaction:** a sharp, sudden price move (a panic sale or a euphoric rally) often partially reverses over the following weeks as the overreaction unwinds — the mechanism behind the short-term *reversal* signal used in this project, and, as it turned out, the single signal that actually worked in the real backtest (Part V).

## 2.5 Extrapolation Bias

**Everyday example:** a company that has grown fast for three years straight is often assumed, informally, to keep growing at a similar pace — even though few businesses sustain any given growth rate indefinitely. People extrapolate recent trends further into the future than the trends themselves justify.

**Market mechanism:** investors chase exciting recent growth stories ("glamour" stocks) and shun boring, currently out-of-favor companies, bidding growth stocks up beyond what their fundamentals justify and value stocks down below theirs. Lakonishok, Shleifer & Vishny's 1994 paper (the founding research behind LSV Asset Management, Part IV) argued this extrapolation bias, not extra risk, explains much of the historical outperformance of value stocks.

## 2.6 Attention and Availability Bias

**Everyday example:** people tend to overestimate the probability of dying in a plane crash relative to a car crash, because plane crashes are rare but vivid and heavily covered by media, so examples come to mind (are "available") more easily than the much more common, less-covered car crash.

**Market mechanism:** retail investors disproportionately buy stocks that have recently grabbed their attention — extreme returns, heavy news or social-media coverage — rather than researching broadly across the market first (Barber & Odean, 2008). This is the central mechanism behind the GameStop episode (Part IV): a stock became a coordinated retail target largely because it was already attention-grabbing, not because of a broad, independent fundamental re-evaluation by thousands of separate investors.

## 2.7 Crowding: Herding at Institutional Scale

**Everyday example:** if every restaurant critic in a city independently decides "fusion cuisine is the next big trend" and reviews accordingly, the city can end up with far more fusion restaurants than genuine demand supports — not because any critic copied another, but because they were all reasoning from similar assumptions and similar data.

**Market mechanism:** many quantitative hedge funds, working independently with different proprietary signals, can still end up holding very similar positions if their signals are built on the same underlying academic factors (value, momentum). Their portfolios look diversified from the inside but are secretly correlated with each other. This is the central mechanism behind the 2007 "Quant Quake" (Part IV): once one large fund was forced to sell, the selling pressure looked, to every other similarly-positioned fund's model, like a signal to sell the same names.

## Part III — Quantitative Methodology, Explained

### 3.1 What is Value-at-Risk, and why does a "calm-regime" model miss the point?

**Plain-language definition:** Value-at-Risk (VaR) at the 99% confidence level answers the question "on a bad-but-not-extreme day (one of the worst 1% of days), roughly how much could this portfolio lose?" It is the financial -industry equivalent of an insurer asking "how much should we expect to pay out in a bad year, one we'd expect to see about once every hundred years?"

**Why a Gaussian (bell-curve) model of VaR is dangerous:** the easiest way to estimate VaR is to assume returns follow a normal (bell-curve) distribution, calibrated on however many years of historical data are available, and to assume the *\*correlations\** between different positions — how much they tend to move together — stay roughly constant. Both assumptions are usually reasonable in ordinary conditions. Both usually fail together, at the worst possible time: during a genuine crisis, returns move by more than a bell curve predicts ("fat tails"), and positions that used to move mostly independently start moving together (correlations spike toward 1), because everyone is reacting to the same underlying fear at once.

#### ANALOGY: TWO HOUSES AND A WILDFIRE

Two houses on the same street, insured separately against fire. Ordinarily, one house catching fire tells an insurer almost nothing about whether the neighboring house will too — fires are close to independent events, so insuring both together looks safe. But a wildfire changes that: it can burn down both houses at once, because now the two risks share a single common cause. A risk model calibrated only on ordinary house fires will badly underestimate the insurer's exposure to a wildfire year, not because the model is bad at estimating ordinary fires, but because it never saw the correlation-raising event in its calibration data. This is exactly the mechanism this project's Part III.3 simulation quantifies for a leveraged trading book.

### 3.2 The Q1 risk simulation: setup

Rather than simply asserting that Gaussian VaR models understate tail risk, this project built a Monte Carlo simulation (*risk\_simulation/fat\_tails\_vs\_normal.py* in the code repository) that computes the underestimation directly. It models a leveraged, six-asset relative-value book (weights summing to roughly 9x gross exposure, loosely evocative of the kind of leveraged spread trades LTCM ran) under two return-generating processes:

- **The model the risk desk believes:** a Gaussian (bell-curve) distribution, with a modest, constant 0.25 pairwise correlation between assets, calibrated on "calm regime" data — exactly the kind of risk model a desk would build from a few years of ordinary historical returns.
- **The process that actually generates the data:** the same calm regime most days, but with a small (1%) daily probability of switching into a "stress regime" where volatility roughly triples and pairwise correlation jumps to 0.92 — combined with fat-tailed (Student-t) shocks in both regimes for realistic extreme-day behavior.

**DECISION & RATIONALE**

This project's sandbox environment had no general internet access when this simulation was built (see Part V for the full story), which ruled out downloading real historical crisis data to measure this effect directly. Rather than skip the question or fake a result, the decision was to build a Monte Carlo simulation that needs no external data at all — every number it reports is a genuine, reproducible output of code that actually ran, even though the underlying scenario is stylized rather than a literal reconstruction of any single historical event. The parameters (0.25 to 0.92 correlation, 3x volatility) were chosen to be qualitatively consistent with what is publicly known about the LTCM episode's shape, not fit to LTCM's actual (never fully disclosed) book — the report is explicit that the resulting multiple should be read as "this is the order of magnitude of the effect," not a precise historical reconstruction.

**3.3 The Q1 results (real, executed numbers)**

| Tail depth | Gaussian (calm-regime) VaR | True (simulated) VaR   | True Expected Shortfall (CVaR) |
|------------|----------------------------|------------------------|--------------------------------|
| 99.0%      | 7.67% of book              | 8.15% of book (1.06x)  | 10.37% of book (1.35x)         |
| 99.9%      | 10.19% of book             | 13.26% of book (1.30x) | 16.83% of book (1.65x)         |

Worst single day across 500,000 simulated trading days: a loss of 58.9% of the book — a scale of move a Gaussian, calm-regime model treats as essentially impossible, but which falls naturally out of a process that occasionally lets volatility and correlation rise together. The gap between the Gaussian model and reality is modest right at the 99% line (that quantile happens to sit near the boundary between regimes in this parameterization) but compounds quickly deeper into the tail — which is itself an important lesson: a risk model can look adequate at the confidence level a regulator or risk committee checks routinely, and still be dangerously wrong exactly where it matters most, a few notches further into the tail.

**3.4 Turning behavioral biases into tradable formulas**

Part II described three biases in plain language. Here is how each becomes a number that can be computed for every stock, every day, and used to rank a universe of companies:

| Signal                 | Formula (plain language)                                                                                           | Bias it targets |
|------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------|
| 12-1 month momentum    | Cumulative return over the trailing ~12 months, excluding the most recent month (to avoid contaminating the trend) | Underreaction   |
| 52-week-high proximity | Current price divided by the highest price reached in the trailing 52 weeks (1.0 = at the high)                    | Overreaction    |
| Short-term reversal    | Negative of the trailing ~1-month return (a recent loser scores positively, betting on a bounce)                   | Overreaction    |

A **composite score** blends all three by converting each to a cross-sectional z-score (how many standard deviations a stock is above or below the average stock, on that signal, on that day) and averaging — the intended purpose being a single number that combines all three biases into one rank. Part V explains why this turned out to be a more dangerous simplification than it first appears.

## 3.5 How the backtest actually works

The methodology follows the standard approach used in the academic momentum and reversal literature, so results are comparable to published findings rather than being an artifact of a bespoke design:

- Once a month, every stock in the universe is ranked by its current signal score.
- The strategy goes equally-weighted *long* the top group (the best-ranked names) and equally-weighted *short* the bottom group (the worst-ranked names) — a "long-short" portfolio, so its return reflects the spread between the best and worst names, not the market's overall direction.
- The portfolio is held for one month, then re-ranked and rebalanced.

Three metrics summarize the result:

- **Sharpe ratio:** average return divided by the volatility (bumpiness) of that return, annualized. A Sharpe of 1.0 is generally considered good for a systematic strategy; a Sharpe near 0 means the strategy earned close to nothing per unit of risk taken; a negative Sharpe means it lost money relative to its own risk.
- **Maximum drawdown:** the largest peak-to-trough decline the strategy experienced at any point in the backtest — a measure of the worst single stretch an investor in the strategy would have had to sit through.
- **Turnover and transaction costs:** every rebalance that changes which names are held costs money in practice (bid-ask spread, market impact). The backtest engine tracks how much of the portfolio changes each month (turnover) and deducts a simple cost (basis points per unit of turnover) to report both a "gross" return (before costs) and a "net" return (after costs) — because a strategy that looks good gross and disappears net of realistic costs is a very common way quantitative research misleads itself.

## Part IV — Five Case Studies, in Depth

Each case below follows the same structure: what happened, with a timeline; which behavioral mechanism from Part II explains it, in plain language; and why the case matters for this project's central question.

### 4.1 Long-Term Capital Management (1998)

#### What happened

Long-Term Capital Management was a hedge fund founded in 1994 by John Meriwether, with partners that included Myron Scholes and Robert Merton, who shared the 1997 Nobel Memorial Prize in Economic Sciences for the Black-Scholes-Merton option-pricing framework taught in every finance curriculum. The fund ran highly leveraged convergence/relative-value arbitrage: betting that prices of closely related securities would converge toward their historically "fair" relationship. Individual trades had small expected edges, so the fund used enormous leverage — over 25:1 on its balance sheet, with derivative notional exposure exceeding \$1 trillion — to turn small, statistically reliable-looking edges into a large return.

From 1994-1997 the fund returned roughly 20-40% annually, a track record that built exactly the kind of confidence Part II.3 (overconfidence) describes. Russia's default and rouble devaluation on August 17, 1998 triggered a global "flight to quality": investors worldwide fled toward the safest, most liquid assets (US Treasuries) and away from everything else, regardless of the specific fundamental link between the assets they were selling. Spreads LTCM's model treated as largely independent instead moved together, in the same direction, and kept widening instead of converging. The fund lost about \$1.9 billion in August alone; by September 21, capital had fallen from roughly \$4.7 billion at the start of the year to about \$600 million, against essentially unchanged notional exposure. On September 23, 1998, with the fund days from a disorderly default that regulators feared could force a fire-sale across Wall Street, the Federal Reserve Bank of New York organized (but did not fund) a \$3.625 billion recapitalization by 14 banks in exchange for 90% of the fund's equity.

#### The behavioral mechanism, explained

This is a herding case (Part II.2), but with an important twist: the herding was not LTCM's own. LTCM's individual trades were, in isolation, defensible. What broke the model was that every other market participant started behaving in a correlated way at once — a collective flight to safety that a "rational agents, independent decisions" model does not naturally produce. The lesson generalizes: a model of rational markets is only as good as its assumption that other participants keep behaving independently under stress, and that is precisely the assumption that breaks in a genuine crisis — the exact mechanism quantified in Part III's Q1 simulation.

## Why it matters here

LTCM is the founding case for this entire project: a fund staffed by the people who literally wrote the mathematics of modern option pricing still lost 90% of its capital in six weeks, not because the mathematics was wrong, but because the mathematics assumed away exactly the kind of correlated human behavior that shows up under real stress.

## 4.2 The "Quant Quake" (August 2007) and Amaranth Advisors (2006)

### What happened — Quant Quake

Between roughly August 6-9, 2007, a large number of market-neutral quantitative equity funds — using different individual signals, run by different firms, with no apparent common trigger in company fundamentals — suffered simultaneous, sharply correlated losses over a few trading days, followed by a partial rebound once the selling pressure eased. Several well-known quant strategies lost on the order of 5-10%+ in that single week — an extreme move for supposedly market-neutral, low-volatility books. The standard academic reconstruction (Khandani & Lo, 2007/2011) attributes this to **crowding**: many quant managers had, independently, arrived at similar factor tilts (value, momentum), so their books were far more correlated with each other than any of them individually realized. When one or more large funds were forced to deleverage a quant book quickly (for reasons believed unrelated to those specific positions), the forced selling looked, to every other similarly-built model, like a signal to sell the same names.

### What happened — Amaranth

Amaranth Advisors was a large multi-strategy hedge fund that lost approximately \$6.5 billion in September 2006, almost entirely from concentrated, highly leveraged natural-gas futures and options positions run by a single trader, Brian Hunter. Hunter's core position was a calendar spread — long winter-month natural-gas futures against short summer-month contracts, betting the spread would widen — sized large enough, relative to actual market open interest, that Amaranth's own trading could move the market it was betting on. A mild start to the 2006 hurricane season pushed the spread the wrong way; over roughly a week in mid-September 2006 the fund lost more than half its capital.

### The behavioral mechanisms, explained

Quant Quake is a crowding case (Part II.7): herding among the most sophisticated participants in the market, the group a naive efficient-markets story would expect to be immune to it. Amaranth is an overconfidence case (Part II.3): Hunter's prior profitable natural-gas trades made a concentrated, survivable-only-if-right position look more justified than the fund's risk limits should have allowed.

### Why they matter here

Together, these two cases show that neither sophistication (Quant Quake) nor a genuinely skilled track record (Amaranth) protects against the specific behavioral failure modes this project studies — they just change which failure mode shows up.

## 4.3 The 2008 Financial Crisis and the Gaussian Copula

### What happened

The 2007-2008 crisis is usually told as a subprime mortgage story, but the mechanism that turned a real but containable housing problem into a systemic banking crisis mirrors this project's Q1 simulation almost exactly. Banks packaged mortgage pools into collateralized debt obligations (CDOs), tranching by seniority. A senior tranche's safety depends critically on how correlated the underlying mortgage defaults are: independent defaults make a senior tranche very safe; correlated defaults (a national downturn hitting every region at once) make that protection evaporate. The industry-standard tool for pricing this correlation risk was the Gaussian copula (David X. Li, 2000), which reduced the hard problem of modeling thousands of correlated defaults to a single correlation parameter, estimated from a history drawn almost entirely from a multi-decade housing boom during which national home prices had never fallen.

When US home prices began falling nationally in 2006-2007 — a regime the calibration window had essentially no examples of — mortgage defaults stopped being independent and started moving together, for the same underlying reason, in every region at once. Senior tranches priced as extremely safe took losses far larger and faster than the model implied, and because these instruments were held throughout the banking system, the losses hit systemically important institutions almost simultaneously (Bear Stearns, March 2008; Lehman Brothers' bankruptcy, September 15, 2008; AIG's roughly \$182 billion federal rescue, driven largely by credit default swaps written against these same instruments).

### The behavioral mechanism, explained

This is structurally the identical mistake as a calm-regime Gaussian VaR model (Part III.1's wildfire analogy): a correlation parameter estimated from a period that never included the correlation-raising regime, and therefore blind to how fast that correlation could rise once it arrived. Two behavioral/institutional layers sit on top of the pure model failure: **herding into a shared industry model** (using it was a competitive necessity even for desks that suspected it understated risk — a rational individual response to career/franchise risk producing an irrational collective outcome), and a **correlated trust collapse** once losses started (nobody could tell who was exposed to how much bad paper, so short-term lenders pulled back from everyone at once — the same flight-to-quality mechanism as LTCM, at systemic scale).

### Why it matters here

This is the largest-scale version, in modern financial history, of the pattern this project's Q1 simulation makes explicit in miniature: a model that is locally reasonable, calibrated on real data, and wrong specifically about how correlation behaves in the rare regime that does the damage.

## 4.4 GameStop and the "Meme Stock" Squeeze (January 2021)

### What happened

Through 2020, several large hedge funds, most prominently Melvin Capital, held large short positions in GameStop (GME), a struggling brick-and-mortar video-game retailer, on a conventional view that the business was in structural decline. Starting in mid-to-late January 2021, a large, coordinated wave of retail buying — organized largely on the Reddit forum r/wallstreetbets — pushed the share price up by roughly an order of magnitude in a few weeks. Two mechanisms compounded pure buying pressure: a **short squeeze** (as the price rose, short sellers faced margin calls and were forced to buy shares to close losing positions, pushing the price up further) and a **gamma squeeze** (heavy buying of short-dated call options forced the market-makers who sold those calls to buy the underlying stock to stay hedged, an amplifying loop running through market structure rather than anyone's fundamental view). Melvin Capital's short position lost the fund a very large share of its capital in January 2021 and received a \$2.75 billion capital injection from Citadel and Point72 on January 25, 2021. Robinhood and other brokerages restricted buying of GameStop and similar stocks on January 28, 2021, citing clearinghouse deposit requirements — a decision that triggered US congressional hearings in February 2021.

### The behavioral mechanisms, explained

This case runs the *opposite* direction from the others: retail herding (Part II.2, at internet speed and scale) overwhelming institutional positioning, amplified by attention-driven trading (Part II.6 — GameStop became a target largely because it was already attention-grabbing, consistent with Barber & Odean's research on individual investors disproportionately buying stocks that recently grabbed their attention) and by the gamified design of mobile-first trading apps.

### Why it matters here

GameStop is included specifically to prevent this project's thesis from collapsing into "institutions are rational, retail is irrational" — a genuinely coordinated crowd can overwhelm sophisticated positioning just as forcefully as institutional herding overwhelmed LTCM's models, just pointed in the opposite direction.

## 4.5 The Other Side: Funds Built to Exploit These Biases

Every case above is a story about a bias causing a loss. These three firms are the counter-evidence: specific, named biases treated as a persistent, exploitable source of return.

### Fuller & Thaler Asset Management

Founded in 1993 by Richard Thaler — awarded the 2017 Nobel Memorial Prize in Economic Sciences for work on how limited rationality affects economic decisions — together with Russell Fuller. Their signature mechanism is closely related to Part II.4's underreaction concept: post-earnings-announcement drift, the well-documented finding that stock prices keep drifting in the direction of an earnings surprise for weeks after the announcement, because analysts anchor on their pre-announcement estimates and update too slowly. Their flagship retail vehicle has long traded as "Undiscovered Managers Behavioral Value" (ticker UBVLX).

## LSV Asset Management

Founded in 1994 by academic economists Josef Lakonishok, Andrei Shleifer, and Robert Vishny, whose own research (Lakonishok, Shleifer & Vishny, *Journal of Finance* 49(5), 1994, pp. 1541-1578) argued that much of value stocks' historical outperformance over "glamour" growth stocks is explained by investors extrapolating recent growth too far into the future (Part II.5), not by value stocks simply being riskier.

## AQR Capital Management

Founded in 1998 by Cliff Asness, whose University of Chicago PhD dissertation (under advisors including Eugene Fama, himself closely associated with the efficient markets hypothesis) was on momentum in stock returns. AQR treats behavioral anomalies as one input among several (alongside risk-based explanations), run with heavy quantitative discipline — a useful middle case between Fuller & Thaler's purely behavioral framing and a pure risk-based factor story.

The working assumption behind all three — and behind this project's signal library in Part III — is that specific, named biases produce persistent, not merely historical, mispricing. Part V.2's empirical result tests that assumption directly, with a real, non-cherry-picked outcome.

## Part V — Project Journal: Milestones, Decisions, Results

This part narrates the project as it actually happened, in order, including the constraints that forced decisions and the results that changed the project's direction. Every decision below is stated together with the reasoning behind it, per this guide's brief: nothing is presented as self-evidently correct without the "why."

### 5.1 Scoping the project

The project started from a broad brief: research the role of human irrationality in markets, with concrete applications to finance and business. Before writing anything, the scope was narrowed by asking four questions: what output format was wanted, which behavioral angle to anchor on, what methodology/evidence to use, and what concrete impact the research should produce.

#### DECISION & RATIONALE

The answers chosen were: a **quantitative research project with runnable code** (not a pure literature review); anchored on **both** failed "rational market" bets (LTCM-style) **and** behavioral funds that exploit bias; using **public market data plus case-study comparison**; aimed at producing an **investment framework, risk-management lessons, and a business idea**. This combination was chosen specifically because it forces the project to be falsifiable in both directions at once (Part I.3) rather than only telling the flattering half of the story (either "markets are irrational, here's how to profit" or "here's why smart people fail," without testing either claim against real numbers).

### 5.2 Milestone 1 — Building the pipeline under a hard constraint

The first concrete obstacle appeared immediately: the development environment's outbound network access is restricted by an egress proxy to a specific allowlist. Direct testing (curl requests to each host, and reading the proxy's own status endpoint, which logs connection failures) confirmed that Yahoo Finance, Stooq, SEC EDGAR, and FRED — every conventional free market-data source — were all blocked with a 403 policy denial.

### DECISION & RATIONALE

The tempting shortcut here would have been to generate synthetic "market-looking" data and present backtest results computed on it as if they were real findings — exactly the kind of model-dressed-as-reality mistake this project's whole thesis warns against. The decision instead was to build the complete, real pipeline (data loader, signal library, backtest engine) as production-quality code, but to sharply separate two categories of claim: results that were **actually computed by running code in this environment** (which get stated as fact) versus code that is **complete and believed correct but not yet exercised on real data** (which gets stated as an explicit, flagged limitation, never as a finding). The one part of the research that needed no external data at all — the Q1 tail-risk simulation (Part III.2-3.3) — was built first specifically because it could produce a genuine, real result despite the network constraint.

Milestone 1 shipped: the research design document, the full signal and backtest codebase (validated only against synthetic test fixtures at this point, with that limitation stated explicitly), the Q1 risk simulation with its real executed numbers, three initial case studies (LTCM, Quant Quake / Amaranth, the behavioral funds), and a first version of the practical framework document.

### PROCESS DECISION

At the end of milestone 1, a new working practice was established at the user's request: after every milestone, provide a recap of the reasoning behind what was built, followed by explicit questions checking whether the project's focus was still correctly calibrated to what was actually wanted. This is why milestones 2 and 3 below each open with what the previous recap's questions surfaced.

## 5.3 Milestone 2 — Real data, and a result that changed the project

The milestone-1 recap asked, among other things, how to handle the no-internet constraint going forward. The chosen answer was to actively search for a reachable data source rather than accept the limitation as final.

### DECISION & RATIONALE

Rather than assume every external host was blocked, a range of unrelated hosts was tested directly — not just financial-data providers but also generic ones (a plain package index, a generic search engine). This showed the block was not "no internet at all" but a specific allowlist: *api.github.com* and *raw.githubusercontent.com* were reachable while general financial-data hosts and even a plain general-purpose website were not. That distinction mattered: it meant GitHub-hosted datasets were a real option, not a dead end.

A web search for GitHub-hosted multi-ticker daily price datasets surfaced a real candidate: a community-uploaded CSV of 63 raw ticker symbols covering NSE (India) large/mid-cap companies from 2000 to 2021. Before using it, the data was checked for basic quality (no duplicate date/symbol rows, no zero or negative prices) and then for a subtler issue: several apparent "different companies" turned out,

on inspection of their trading date ranges, to be the *same* company under a former ticker symbol.

#### DECISION & RATIONALE

The evidence for this was date-range contiguity: symbol A's last trading date and symbol B's first trading date lined up with no overlap and no gap, for 14 separate symbol pairs (or longer chains) — the exact signature of an NSE ticker rename. Each chain was then cross-checked against publicly known Indian corporate history (for example, "TELCO" was Tata Motors' ticker before a rename; "SESAGOA" became "SSLT" then "VEDL" as Sesa Goa became Sesa Sterlite became Vedanta Ltd) before merging. One pair, HDFC and HDFCBANK, was deliberately *not* merged, because the two were genuinely separate listed companies for the entire sample period (they only merged in 2023, after the data ends). This is the kind of judgment call that is easy to get wrong silently in a backtest — merging two unrelated companies, or failing to merge one company's own history — so it was done from evidence, and locked in afterward with a dedicated automated test (visible in the code repository's `tests/test_loaders.py`) rather than left as a one-off manual step.

With 48 clean, continuous company series covering 21 years, the full signal-and-backtest pipeline was run for the first time on real market data.

| Signal                   | Split     | Gross annual return | Gross Sharpe | Net Sharpe | Max drawdown |
|--------------------------|-----------|---------------------|--------------|------------|--------------|
| Composite (all 3)        | deciles   | -3.59%              | 0.01         | -0.03      | -74.3%       |
| Composite (all 3)        | quintiles | -1.33%              | 0.05         | 0.00       | -63.2%       |
| 12-1 momentum only       | quintiles | -1.73%              | 0.04         | 0.01       | -60.8%       |
| 52-week-high only        | quintiles | -13.71%             | -0.51        | -0.54      | -97.1%       |
| Short-term reversal only | quintiles | +4.46%              | 0.30         | 0.22       | -64.1%       |

The composite score alone would have supported a bland, roughly-correct-sounding conclusion: "no real edge here." Decomposing it into its three parts told a much more informative and, in one case, misleading story: the 52-week-high (anchoring) signal was actively destructive, losing money persistently and suffering a 97% drawdown, while the short-term reversal (overreaction) signal *looked* like a genuine, positive, cost-adjusted edge. Averaging them together hid both facts — and Part V.9 (Milestone 8) later found the reversal side of that story didn't survive the same scrutiny the losing side got.

A follow-up check tested whether the anchoring signal's damage was simply an artifact of the 2008 and 2020 crashes (both periods where a "buy recent winners" strategy is known from the academic literature to be exposed to sharp reversals — "momentum crash risk"). Excluding both crash windows entirely, the strategy's Sharpe ratio outside them was still -0.46, and its cumulative loss *excluding* the crashes (-90%) was actually larger than its loss *during* them (-53%) — ruling out "it only lost money in the two crashes" as the explanation, and leaving the true cause an open question flagged honestly rather than resolved with a plausible-sounding guess. (Part V.4 below picks this exact question back up.)

**DECISION & RATIONALE**

The milestone-2 recap surfaced this same lesson back to the practice guiding the project itself: a mediocre-looking blended result should always be decomposed before being trusted, a rule now stated explicitly in the practical framework (Part VI.1). The recap also asked whether tying the eventual business idea to this repository's own existing software platform (a private-equity deal-screening tool the code happened to live alongside) was the right framing. The answer was no — the business idea was rewritten from scratch as a standalone analytics service (Part VI.3), decoupled from any specific existing platform, per that explicit feedback.

## 5.4 Milestone 3 — Replication, and a rejected hypothesis

The milestone-2 recap's questions produced three directions: search for a second, independent market's data to check whether the NSE findings replicate elsewhere; investigate *why* the 52-week-high signal lost money, specifically testing whether it was confounded with a value/growth effect; and continue expanding the codebase rather than wrapping up. Producing this guide itself was folded into the same milestone, at the user's explicit request to run it in parallel with the quantitative work rather than after it — which is why an earlier draft of this section reported milestone 3 as in-progress; both investigations below have since completed.

### Finding a second market

A GitHub-hosted mirror of the well-known Kaggle "Huge Stock Market Dataset" (one CSV per US-listed ticker) was located and verified: 30 US large-cap names, the same tickers as the project's original default universe, each starting at its own listing date and running through 2017-11-10 (the dataset's last snapshot; this is a historical replication check, not a live feed).

| Signal                   | Split     | Gross annual return | Gross Sharpe | Net Sharpe | Max drawdown |
|--------------------------|-----------|---------------------|--------------|------------|--------------|
| Composite (all 3)        | quintiles | +2.99%              | 0.25         | 0.21       | -79.6%       |
| 12-1 momentum only       | quintiles | +5.72%              | 0.34         | 0.32       | -88.4%       |
| 52-week-high only        | quintiles | -13.98%             | -0.29        | -0.32      | -99.9%       |
| Short-term reversal only | quintiles | -0.68%              | 0.13         | 0.07       | -83.0%       |

The result was not a clean confirmation of the NSE finding — it was more informative than that. **Momentum and reversal flipped which one looked real:** reversal was NSE's edge and the US market's near-zero result; momentum was NSE's near-zero result and the US market's real, positive, cost-adjusted edge (Sharpe 0.32 net). Only one thing held up unchanged across both very different markets and eras: **the 52-week-high signal lost money in both.** (Part V.9, Milestone 8, later put both sides of this flip through a beta-adjusted significance test: the US momentum side held up and then some; the NSE reversal side did not.)

**DECISION & RATIONALE**

A single-market backtest result is provisional by default. Had this project stopped after milestone 2 (NSE only), it would have reported "momentum is dead here, reversal is the real edge" — the opposite of what the US data alone would have suggested. Neither market alone is the right answer; the honest finding is that anomaly presence is universe-dependent for momentum and reversal, while the 52-week-high signal's damage is not.

**Testing the value/growth confound — and rejecting it**

The leading candidate explanation for the 52-week-high signal's losses, carried over from milestone 2, was that India's two-decade secular bull market could make "short the names far from their 52-week high" equivalent to "short cheap, beaten-down names" that then mean-revert upward — a value effect masquerading as an anchoring effect. This was tested directly: a simple price-only value proxy (current price divided by its own trailing ~3-year average) was built, its cross-sectional correlation with the raw 52-week-high score was measured, and the 52-week-high signal was then **orthogonalized** against it — a date-by-date cross-sectional regression that mathematically removes the shared value component, leaving only the part of the signal that isn't explained by it.

| Market             | Correlation (signal, value proxy) | Raw signal net Sharpe | Orthogonalized net Sharpe |
|--------------------|-----------------------------------|-----------------------|---------------------------|
| NSE (India)        | 0.63                              | -0.54                 | -0.73                     |
| US (Kaggle mirror) | 0.44                              | -0.32                 | -0.49                     |

The two signals genuinely are correlated (0.44-0.63) — a stock near its 52-week high does tend to look "expensive" on this proxy. But removing that shared component made performance **worse**, not better, in both markets. If the value/growth confound had been the real explanation, the "purified" signal should have looked neutral or better once the confounding value component was stripped out; instead it got more negative. **The value/growth confound hypothesis is rejected.**

**WHAT THIS DOES AND DOESN'T PROVE**

This is a negative result, and it was reported as one rather than quietly dropped or reframed as a success. Combined with the milestone-2 finding that the loss isn't concentrated in the 2008/2020 crash windows either, two of the three candidate explanations for the 52-week-high result are now ruled out by direct test, not just by argument. The remaining candidate — generic momentum-crash risk, a documented property of "long recent winners" strategies in general — is the best-supported explanation left standing, but it has not itself been directly tested (that would mean checking whether the losses cluster specifically in high-volatility periods). The project states this as narrowed, not solved.

**5.5 Milestone 4 — Testing momentum-crash risk directly**

**READ THIS FIRST**

**Update (Section 5.6, Milestone 5): the formal statistical test does NOT confirm the regime-conditioning mechanism this section describes.** This section is kept as written at the time — the pattern-matching genuinely looked this compelling — but Section 5.6 walks it back with real numbers. Read that section before treating anything below as confirmed.

The one remaining candidate explanation from milestone 3 — generic momentum-crash risk (Daniel & Moskowitz, "Momentum Crashes," 2016; see Part II.4 and Part IV.1) — was tested directly at the user's request. The theory's specific prediction: a "long recent winners, short recent losers" strategy is exposed to severe losses when realized market volatility is elevated and the market has recently been in a downturn, because the *short* leg (recent losers, typically higher-beta) can rebound sharply in a recovery — and the damage should concentrate specifically in that short leg, not be spread evenly across both sides of the trade.

Two things were built to test this properly rather than just plausibly: two regime indicators from the same price data (a realized-volatility tercile and a trailing-12-month bull/bear flag, both lagged one day to avoid look-ahead), and an extension to the backtest engine itself so it reports the long leg's and short leg's profit-and-loss contributions *separately*, not just their combined total — needed to check the mechanism, not only the timing.

| Evidence                                       | NSE (India)             | US (Kaggle mirror)      |
|------------------------------------------------|-------------------------|-------------------------|
| Sharpe vs. realized-vol tercile (Low/Mid/High) | -0.22 / -0.41 / -0.78   | -0.11 / -0.27 / -0.43   |
| Sharpe, bull vs. bear trailing-12m state       | -0.27 vs. -1.06         | -0.15 vs. -1.10         |
| Worst bucket: bear + high-vol                  | -1.07 Sharpe, -38% ann. | -1.26 Sharpe, -61% ann. |
| Return skewness, combined long-short           | -0.66                   | -1.32                   |

All four signatures point the same way, in both markets. The leg decomposition confirms the mechanism, not just correlated timing: the **long leg (buying stocks near their 52-week high) is a genuinely positive, standalone strategy** in both markets (Sharpe 0.45-1.03 in every regime tested on NSE; 0.65-1.20 outside bear markets in the US) — the anchoring/underreaction thesis actually works on the long side. **The short leg is the entire problem:** negative in every single regime bucket in both markets, and its losses are specifically what swells in the bear-plus-high-volatility bucket (NSE: -19% to -49% annualized from calmest to worst regime; US: -13% to -62%).

**HOW HARD TO LEAN ON THIS, AS UNDERSTOOD AT THE TIME**

This is pattern-matching against a well-documented mechanism using real data across two independent markets, not a formal significance test — no statistical t-tests were run on the regime differences, and the volatility-tercile cutoffs were computed over each dataset's full history rather than a rolling window a live system would need. With that caveat stated plainly, at this point in the project four independent, mutually consistent signatures looked like about as strong a case as this kind of historical analysis could make short of formal statistics — which is exactly what Section 5.6 adds, and exactly what changes the conclusion.

## 5.6 Milestone 5 — Formally testing momentum-crash risk

The user explicitly asked for the pattern-matching in Section 5.5 to become "a statistical test with proven reliability" before opening a pull request. Two upgrades were made (*investigations/momentum\_crash\_significance.py*): the volatility tercile threshold now uses an **expanding window** (each day's "high vol" label uses only data through the previous day, fixing the one look-ahead caveat Section 5.5 flagged), and strategy returns are regressed on the regime indicators using **Newey-West (HAC) standard errors** — the standard correction for serial correlation from monthly-rebalanced, overlapping holding periods, the same style of correction the original Daniel & Moskowitz paper itself uses. Tests were run at both daily frequency (HAC lag 21) and monthly frequency (one return per rebalance, HAC lag 6, matching the paper's own testing frequency).

| Return series                            | Daily: high_vol x bear | Monthly: bear | Monthly: vol_rank   |
|------------------------------------------|------------------------|---------------|---------------------|
| NSE short leg (theory's predicted locus) | p=0.57                 | p=0.67        | p=0.09 (wrong sign) |
| NSE long leg                             | p=0.94                 | p=0.0007      | p=0.0046            |
| US short leg (theory's predicted locus)  | p=0.65                 | p=0.34        | p=0.61              |
| US long leg                              | p=0.15                 | p<0.0001      | p=0.0139            |

**The specific regime-conditioning mechanism does not survive formal testing.** The short leg — the one momentum-crash theory specifically predicts should blow up in high-volatility, bear-market regimes — shows no statistically significant regime-conditioning in either market, at either frequency, in any specification tested. NSE's short leg even has a wrong-signed coefficient at the 10% level (losses *shrinking*, not growing, as volatility rises) — evidence against the hypothesis in that specific cut, not for it. The regime effects that *are* highly significant ( $p < 0.01$ , every specification) show up in the long leg instead — a far more mundane explanation (a long-biased position carries positive market-beta exposure and underperforms generally in bear markets) than the specific short-squeeze mechanism the hypothesis describes.

**WHAT SURVIVES AND WHAT DOESN'T, AS UNDERSTOOD AT THIS POINT**

What remains statistically ironclad, every specification, both markets, both frequencies: the long leg's average return is significantly positive and the short leg's is significantly negative — the decomposition itself, not the proposed explanation for it. With roughly 24 regime-effect coefficients tested across both frequencies and markets, seeing 2-4 marginally significant results at the 5-10% level is within what pure chance produces under a true null — not meaningfully more than a false-positive rate. As understood at this point, the long-only recommendation still stood — but the project cannot claim to know *why* the short leg fails. That remained genuinely open, and Section 5.7 is where it actually gets resolved, one more check further.

## 5.7 Milestone 6 — Is it just beta?

Milestone 5 left one basic check untried: does the long leg and the short leg simply have different, uncontrolled exposure to the market itself? An equal-weighted decile long-short book makes no attempt to match the two baskets' market sensitivity. If "losers" (far from the 52-week high) happen to be higher-beta stocks than "winners" (near the high), the strategy carries unintended net market exposure — and in a market with a strongly positive average return over the sample (both NSE and the US mirror returned roughly 19-21% annualized over their windows), that alone would produce exactly the pattern seen, with no anchoring or crash story required.

A standard CAPM-style regression was run (*investigations/short\_leg\_beta.py*):  $\text{leg return} = \alpha + \beta \times \text{market return}$ , using the same equal-weighted market proxy and the same Newey-West correction as Section 5.6, at daily and monthly frequency, for the long leg, the short leg, and the combined book.

|                       | NSE daily       | NSE monthly     | US daily        | US monthly      |
|-----------------------|-----------------|-----------------|-----------------|-----------------|
| Long leg beta (p)     | +0.84 (p<.0001) | +0.80 (p<.0001) | +0.82 (p<.0001) | +0.75 (p<.0001) |
| Short leg beta (p)    | -1.16 (p<.0001) | -1.19 (p<.0001) | -1.39 (p<.0001) | -1.40 (p<.0001) |
| Combined net beta (p) | -0.32 (p<.0001) | -0.40 (p=.0007) | -0.57 (p<.0001) | -0.70 (p<.0001) |
| Long leg alpha (p)    | p=0.60          | p=0.61          | p=0.76          | p=0.35          |
| Short leg alpha (p)   | p=0.15          | p=0.65          | p=0.72          | p=0.95          |
| Combined alpha (p)    | p=0.25          | p=0.42          | p=0.69          | p=0.41          |

**This is the answer.** All 12 beta coefficients (both legs plus the combined book, both markets, both frequencies) are highly significant (p<0.001, all but one below p=0.0001). All 12 alphas are statistically indistinguishable from zero (p from 0.15 to 0.95). The "losers" basket consistently carries a larger-magnitude beta than the "winners" basket — consistent with the momentum literature's own description of loser-leg stocks as higher-beta — but the consequence here is purely mechanical: **the combined book is not market-neutral. It carries a significant, unintended net short-beta position (-0.32 to -0.70), and being structurally short a market that returned ~20% a year is a completely sufficient explanation for the losses documented in every milestone above — no anchoring bias,**

**no crash risk, no value confound required.**

#### WHAT THIS CHANGES

This changes the practical recommendation from Sections 5.5-5.6, not just adds a footnote. "Trade the signal long-only" rested on the long leg's raw average return being significantly positive. That return is real, but this test shows it's consistent with simply holding an 0.75-0.84-beta long position in a rising market, with **no significant standalone alpha** once that exposure is controlled for. There is currently no evidence, anywhere in this project, of genuine stock-selection skill in the 52-week-high signal, in either direction, once market beta is properly accounted for. The honest recommendation: this signal's long leg is a reasonable lower-beta way to stay invested, not a demonstrated behavioral edge. Testing for real, beta-independent skill would require explicitly beta-neutralizing both legs and re-running this same regression on the hedged residual — done next, in Section 5.8.

#### THE LESSON, AGAIN

The real methodological lesson: three milestones (2, 3, 5) chased increasingly sophisticated behavioral and statistical explanations — crash windows, a value/growth confound, formal regime-conditioning significance tests — before checking the single most basic hygiene check for any long-short equity backtest: whether the two legs are beta-matched. They weren't, and that omission alone explains everything the more exotic hypotheses were built to explain. Order hypothesis tests from cheapest and most mechanical to most exotic, not the reverse.

## 5.8 Milestone 7 — Does real alpha survive an actual beta hedge?

Milestone 6 used one full-sample regression coefficient to estimate beta and asked whether the leftover average return was significantly different from zero. That is a legitimate test, but it assumes constant market exposure across 20+ years, which real betas don't have. This milestone builds and tests the real thing (*investigations/beta\_hedged\_backtest.py*): at every monthly rebalance, beta is re-estimated from only the **trailing 252 trading days** (roughly one year) of the combined book's own history — strictly out-of-sample, never using data from the period being hedged — and that beta is used to hedge the *next* month's daily returns. This mirrors how a real fund would operationally hedge: periodic re-estimation applied forward, no look-ahead.

|                                         | NSE (India)            | US (Kaggle mirror)     |
|-----------------------------------------|------------------------|------------------------|
| Days covered (needs 252d history)       | 5,142                  | 11,417                 |
| Rolling hedge beta: mean (std)          | -0.26 (0.31)           | -0.44 (0.50)           |
| Correlation of hedged returns w/ market | -0.07                  | -0.02                  |
| Unhedged ann. return / vol / Sharpe     | -14.3% / 23.9% / -0.53 | -14.7% / 33.5% / -0.30 |
| Hedged ann. return / vol / Sharpe       | -7.1% / 22.7% / -0.21  | -7.5% / 31.3% / -0.09  |

|                                    | NSE (India) | US (Kaggle mirror) |
|------------------------------------|-------------|--------------------|
| Hedged daily return significant?   | p=0.35      | p=0.45             |
| Hedged monthly return significant? | p=0.42      | p=0.17             |

**Independent confirmation of Milestone 6, by a stronger method.** The rolling hedge substantially cuts the strategy's correlation with the market (from strongly negative down to -0.02 to -0.07) and **roughly halves the annualized loss** in both markets — consistent with a large share of the original loss being mechanical beta exposure, not something specific to the "losers" basket. But even after this real, out-of-sample hedge, **the residual return is not statistically distinguishable from zero in either market, at either frequency** (p between 0.17 and 0.45). No significant alpha, positive or negative, survives once market exposure is genuinely — not just statistically — removed.

#### TWO HONEST CAVEATS

Two honest caveats. First, the hedge is imperfect: residual correlation isn't exactly zero, and the rolling beta itself is quite unstable over time (its standard deviation is comparable to or larger than its mean in both markets) — a real hedging program would need frequent rebalancing and would incur transaction costs this script doesn't model. Second, a real, textbook bug was caught and fixed before any of these numbers were reported: an early version estimated beta as  $np.cov(...) / np.var(...)$ , but NumPy's `cov` defaults to a different degrees-of-freedom convention (`ddof=1`) than its `var` (`ddof=0`) — silently inflating every beta estimate by roughly  $n/(n-1)$  (about 0.4% here, immaterial to the conclusion, but a real bug regardless). Fixed by using an OLS slope instead, and locked in with a synthetic-fixture test asserting exact beta recovery on a zero-noise series.

## 5.9 Milestone 8 — Does the reversal edge survive a beta check too?

Every beta check so far (Sections 5.7-5.8) was run on the 52-week-high signal — the one that *lost* money. Short-term reversal was this project's one positive empirical finding (Section 5.3: "the only signal that actually worked, gross and net of costs," Sharpe 0.22 net on NSE), and it was never re-examined for the exact same uncontrolled-beta artifact that turned out to fully explain the 52-week-high signal's losses. This milestone applies the identical CAPM-style test (*investigations/momentum\_reversal\_beta.py*) to **both** remaining signals — 12-1 momentum and short-term reversal — long leg, short leg, and combined book, both markets, both frequencies: 24 alpha tests in total.

**Short-term reversal's "edge" does not survive.** Not one of its 12 alpha tests is significant at conventional levels (all  $p > 0.10$ , most  $p > 0.2$ ). The positive Sharpe reported all the way back in Section 5.3 was — like the 52-week-high signal's loss — a mix of market-beta exposure and noise, not a demonstrated stock-selection edge. **This project's one previously "positive" empirical result is retracted along with the negative one.**

**12-1 momentum tells a genuinely different story.** On NSE, momentum shows essentially no significant alpha either (one marginal hit at the 10% level). But on the **US mirror, momentum's long leg and**

**combined book show real, statistically robust, largely beta-independent alpha:**

|                            | US long leg           | US combined book                        |
|----------------------------|-----------------------|-----------------------------------------|
| Daily alpha (annualized)   | +8.1%/yr, $p < .0001$ | +12.1%/yr, $p = .004$                   |
| Monthly alpha (annualized) | +9.2%/yr, $p < .0001$ | +15.3%/yr, $p < .0001$                  |
| Beta                       | ~+1.0 to +1.1         | ~-0.05 to -0.24, mostly not significant |

The combined long-short book's beta is close to zero and, at daily frequency, not statistically different from zero ( $p = 0.38$ ) — this book is close to genuinely market-neutral **and** has a large, highly significant positive average return. Unlike the marginal, scattered hits dismissed as noise in Section 5.6's multiple-testing discussion, this is a **coherent cluster**: the same signal, the same market, the same direction, significant at the 1% level or better across both frequencies and both the long leg and the combined book — a qualitatively different, much less noise-like pattern than an isolated  $p \approx 0.03$  hit.

#### HOW HARD TO LEAN ON THIS, AS UNDERSTOOD AT THE TIME

This is the strongest, most credible finding in the entire project — but it is one market (the US Kaggle mirror, snapshot ending 2017-11-10), it does not replicate on NSE, and it has not been checked for the specific decay risk this guide's own limitations list has flagged since Part III: momentum was published by Jegadeesh & Titman in 1993, and momentum's premium is well documented in the literature to have weakened somewhat post-publication. Whether this specific alpha holds up in a pre-1994 vs. post-1994 sub-sample split has **not yet been tested** — checked next.

## 5.10 Milestone 9 — Has US momentum's alpha decayed since publication?

Jegadeesh & Titman published the 12-1 momentum anomaly in the *Journal of Finance* in March 1993. A large literature (notably McLean & Pontiff, "Does Academic Research Destroy Stock Return Predictability?", *Journal of Finance*, 2016) documents that anomaly returns tend to shrink — by roughly 26% after working-paper circulation and ~58% on average after formal publication — once traders can crowd into a known effect. This milestone (*investigations/momentum\_publication\_decay.py*) splits Section 5.9's long leg and combined-book regressions at **1994-01-01** (~1 year after publication) and re-runs each half separately: 251 pre-1994 rebalances vs. 286 post-1994, out of 537 total.

|                      | Long leg, pre-1994     | Long leg, post-1994  | Combined, pre-1994     | Combined, post-1994         |
|----------------------|------------------------|----------------------|------------------------|-----------------------------|
| Daily alpha (ann.)   | +9.8%/yr, $p = .003$   | +6.5%/yr, $p = .032$ | +16.1%/yr, $p = .011$  | +8.6%/yr, $p = .116$ (n.s.) |
| Monthly alpha (ann.) | +11.4%/yr, $p = .0004$ | +7.1%/yr, $p = .041$ | +18.6%/yr, $p = .0008$ | +13.1%/yr, $p = .035$       |

**Real, partial decay — exactly the textbook pattern, not full disappearance.** Every cut shows the alpha shrinking after 1994: the long leg's daily alpha falls by roughly a third (9.8% to 6.5%/yr) and its monthly alpha by roughly two-fifths (11.4% to 7.1%/yr); the combined book decays by 30-47% depending

on frequency. That magnitude lines up closely with McLean & Pontiff's average post-publication effect across anomalies generally — this isn't an unusually large or suspicious decay, it's the expected one. **The alpha survives in three of four cuts** (long leg, both frequencies; combined book, monthly) but **loses statistical significance in one** (combined book, daily,  $p=0.116$ ). The combined book's beta also drifts from indistinguishable-from-zero pre-1994 (genuinely market-neutral) to weakly negative post-1994 — a secondary sign that the strategy's risk profile itself shifted once the anomaly became public knowledge.

#### THE HONEST, FINAL-SIZED NUMBER

This project's one durable finding is now stated more precisely: momentum's long leg (and, less robustly, the market-neutral combined book) continued to carry statistically significant, economically smaller alpha through 2017 — not an un-decayed anomaly. That is a real, still-standing finding — three of four regression cuts remain significant in the post-1994 half alone, more than 20 years after publication — but a meaningfully weaker one than Section 5.9's full-sample numbers suggested on their own. **Size any position on the post-decay number, not the full-sample one** — the pre-publication half of the sample describes a market that no longer exists.

#### UPDATE FROM SECTION 5.11

Section 5.11 re-tested this exact post-1994 number with an actual out-of-sample hedge instead of an in-sample regression, and found it does not survive. Read the box below as the current, superseding answer to "is this decayed number real".

## 5.11 Milestone 10 — Does the post-1994 alpha survive an actual out-of-sample hedge?

Section 5.10's decay split is a real improvement over a full-sample average, but it has the same limitation Section 5.7's beta regression had before Section 5.8 closed the loop: it fits one beta **in-sample**, using the whole post-1994 sub-sample's own data, then asks whether the average residual differs from zero. A real fund cannot do that — it has to estimate beta from only trailing history and hedge forward. This milestone (*investigations/momentum\_hedged\_decay\_backtest.py*) reuses Section 5.8's rolling, out-of-sample beta hedge (re-estimated every monthly rebalance from only the preceding 252 trading days, applied forward, never looking ahead) on the momentum long leg and combined book, then applies the identical 1994-01-01 split to the resulting **hedged** daily return series.

|                     | Long leg, pre-1994 | Long leg, post-1994 | Combined, pre-1994 | Combined, post-1994 |
|---------------------|--------------------|---------------------|--------------------|---------------------|
| Hedged ann. return  | +9.62%/yr          | +3.29%/yr           | +7.93%/yr          | -0.31%/yr           |
| Daily alpha (HAC)   | $p=.0010$          | $p=.1485$ (n.s.)    | $p=.0244$          | $p=.5900$ (n.s.)    |
| Monthly alpha (HAC) | $p=.0017$          | $p=.1886$ (n.s.)    | $p=.0506$ (n.s.)   | $p=.5049$ (n.s.)    |

**This is a further, sharper correction, not a confirmation of Section 5.10's "survives in three of four cuts" framing.** Once beta is estimated the way a real fund would have to estimate it — from trailing data

only, re-hedged every month, never fit on the same period being tested — the post-1994 alpha is **not statistically distinguishable from zero in either leg**, at either frequency. The combined book's hedged post-1994 average return is outright negative. Pre-1994, the same methodology strongly confirms alpha in both legs ( $p \leq 0.025$  in three of four cuts), so the hedge itself isn't simply too noisy to detect a real effect when one is present — it detects it clearly pre-1994 and finds nothing post-1994. The gap between Section 5.10's "three of four cuts survive" and this section's "none survive" is entirely explained by the in-sample vs. out-of-sample distinction: an in-sample regression can fit the specific quirks of the post-1994 data it's being tested against, while a rolling hedge estimated only from prior data cannot.

#### THE OUT-OF-SAMPLE-CONFIRMED ANSWER, AS FAR AS THIS SECTION TOOK IT

US 12-1 momentum's alpha was real and strong before 1994 and has **not** been demonstrated to survive, in a form an actual fund could have traded, in the more-than-two-decades since. This project's single most credible candidate for a genuine, durable edge does not clear the bar once tested with the same standard of rigor — an actual rolling out-of-sample hedge, not a static or per-era regression coefficient — that Section 5.8 already established as this project's own required standard. This doesn't mean Sections 5.9 and 5.10's findings were wrong for their samples: the full-sample and pre-1994 alpha are both real and robust. It means the finding cannot currently be sized as a forward-looking edge without further work.

#### UPDATE FROM SECTION 5.12

Section 5.12 investigated this null result in depth, at explicit request, rather than stopping here: is it really just the 2009 momentum crash, or an artifact of the hedge's rolling window? Neither. Read on for the deeper, still-standing answer.

## 5.12 Milestone 11 — Is the post-1994 null result a crash artifact, a hedge artifact, or genuine decay?

Section 5.11 left two objections explicitly untested: is the post-1994 null result concentrated in a specific regime — most plausibly the well-documented 2009 "momentum crash" (Daniel & Moskowitz, 2016, *Review of Financial Studies*, where past losers momentum strategies were underweighting rebounded violently in the 2008-crisis recovery) — or is it an artifact of the specific 252-trading-day rolling hedge window Sections 5.8 and 5.11 happened to use? This milestone (*investigations/momentum\_decay\_regime\_analysis.py*) runs three checks on the post-1994 hedged return series to answer both.

**Check 1 — sub-period breakdown.** Splitting post-1994 into five multi-year eras shows a declining trend, not a single bad episode with a recovery: the long leg's hedged excess return runs +6.5%, +10.5%, +3.4%, -3.7%, then **+0.1%** in 2010-2017 — the most recent nine years show essentially zero excess return, well after the 2009 crash was over. The combined book shows the same pattern, ending at **-5.1%** in 2010-2017.

**Check 2 — explicit crash-window exclusion (March-August 2009).** The crash window itself was severe — the combined book lost 40.2% cumulatively in those 128 trading days alone. Excluding it: the long leg's daily alpha p-value improves from 0.149 to a borderline 0.095 (the crash meaningfully hurt this leg's result), but the combined book's p-value only improves from 0.590 to 0.334, nowhere near significance. The crash is a real contributing factor for the long leg, but does not explain the combined book's null result, and the continued weakness through 2010-2017 shows the effect is not fully explained by a single 2009 episode either way.

**Check 3 — hedge rolling-window robustness (126d / 252d / 378d).** Re-running the entire hedge with three different lookback windows: pre-1994 alpha is strongly significant at every window tested ( $p \leq 0.017$ , long leg  $p \leq 0.001$ ) and post-1994 alpha is non-significant at every window tested ( $p$  ranges 0.14-0.66 across both legs and all three windows). The null result is not an artifact of the specific 252-day window.

#### THE DEEPER, STILL-STANDING ANSWER

This deeper investigation does not reverse Section 5.11's finding — if anything it strengthens it. The 2009 momentum crash was real and severe and meaningfully affected the long leg's result, but it is not sufficient on its own to explain either leg's post-1994 null result, and the hedge's design choice is not driving it either. **The era-by-era trend (positive and shrinking through the 2000s, negative through the crisis, and still flat or negative for the nine years since) is the signature of genuine, ongoing decay rather than one bad shock the strategy simply hasn't yet recovered from.** A finding that survives a genuine attempt to explain it away deserves more confidence, not less.

#### UPDATE FROM SECTION 5.13

Section 5.13 asks whether this whole pattern is unique to momentum, by applying the same toolkit to the other two US signals. The answer splits: one signal never had genuine alpha at all; the other shows the identical decay signature as momentum.

## 5.13 Milestone 12 — Is "real pre-1994, decayed since" specific to momentum, or market-wide?

Sections 5.10-5.12 built a specific toolkit — a rolling, out-of-sample beta hedge split at 1994-01-01 — and applied it only to momentum, because momentum was the one signal with significant full-sample alpha worth investigating. But the 52-week-high and short-term reversal signals were both declared dead using a *full-sample* beta-adjusted regression (Sections 5.7-5.9), which would hide the exact same pattern found for momentum: real, significant alpha in an early era, averaged down to statistical noise by a decayed later era. This milestone (*investigations/all\_signals\_decay\_analysis.py*) applies the identical pre/post-1994 hedge to **all three** US signals — not to re-answer an already-settled question (NSE momentum's full-sample null was already established in Section 5.9, so re-running the hedge there would add nothing), but to ask a genuinely new one: is the decay pattern unique to momentum, or a broader feature of the US market?

| Signal / leg            | Pre-1994 ann. ret / p | Post-1994 ann. ret / p |
|-------------------------|-----------------------|------------------------|
| 12-1 momentum, long leg | +9.62%/yr, p=.001     | +3.29%/yr, p=.149      |
| 12-1 momentum, combined | +7.93%/yr, p=.024     | -0.31%/yr, p=.590      |
| 52-week-high, long leg  | -0.21%/yr, p=.704     | -2.06%/yr, p=.468      |
| 52-week-high, combined  | -8.46%/yr, p=.776     | -6.68%/yr, p=.401      |
| Reversal, long leg      | +5.31%/yr, p=.046     | +0.25%/yr, p=.629      |
| Reversal, combined      | -0.41%/yr, p=.293     | -0.57%/yr, p=.576      |

**Two genuinely different stories, not one.** The 52-week-high signal shows **no significant alpha in either era, in any leg** — confirming Sections 5.7-5.8's conclusion that this signal never had genuine stock-selection skill in either direction; its full-sample "loss" was uncontrolled beta from the start, not a decayed edge. **Short-term reversal's long leg tells a different story: it shows the identical decay pattern as momentum** — a real, statistically significant pre-1994 alpha (+5.31%/yr, p=0.046) that decays completely to noise post-1994 (+0.25%/yr, p=0.629). This was invisible in Section 5.9's full-sample regression, which averaged the genuine early effect with the decayed later one and correctly found no full-sample significance — but "no full-sample significance" is not the same claim as "never had a genuine edge," and this milestone shows reversal's long leg did.

#### TWO SIGNALS, ONE SHARED STORY

Section 5.9's headline claim — that reversal's "edge" did not survive the same scrutiny applied to the negative one — is accurate for the full-sample regression it ran, but incomplete: reversal's long leg was never pure noise, it was a real, decayed effect exactly analogous to momentum's, just never tested with the era-split methodology that only existed starting at Section 5.10. **The "real pre-1994, decayed since" pattern is not a momentum-specific quirk — it appears in two of this project's three signals' long legs (momentum and reversal) and is absent from the third (52-week-high, which never had genuine alpha at all)**, consistent with a market-wide explanation — the same 1990s-2000s scaling-up of quantitative, cross-sectional equity strategies this project's own case studies (LTCM, the 2007 Quant Quake) already document as having transformed US equity markets over exactly this period — rather than an idiosyncratic property of momentum alone.

## Part VI — From Research to Practice

### 6.1 An investment framework: the Behavioral Mispricing Score

The composite score from Part III.4 is designed to be used as a **tilt or flag layered on top of an existing fundamentals-driven screen**, never as a standalone strategy — the same pattern every fund in Part IV.5 actually follows (they all pair behavioral signals with fundamental discipline, not raw signal-following). A stock that screens well fundamentally *and* sits in the top behavioral decile is a stronger candidate than either signal alone would suggest; a stock flagged as a behavioral "loser" deserves a second look even when its fundamentals look fine.

#### DECISION & RATIONALE

Rule, stated as a direct consequence of Part V.3's finding, not as generic best practice borrowed from a textbook: **never trust a blended score without checking what each of its components does on its own**. The composite here looked mediocre; decomposing it found one component actively losing money with a 97% drawdown and another that at first looked like a genuine, positive, cost-adjusted edge (a claim Part V.9 later retracted). Any score this project's code produces should be reported and validated component-by-component before a blended version of it is trusted for a decision.

**Decompose the legs before discarding a signal — still correct, but the fix isn't "long-only" after all.** The long leg's raw average return is significantly positive, the short leg's significantly negative, in every specification tested (Part V.6). That decomposition rule stands. What it was used to conclude did not: Part V.7 found both legs' significant raw returns are fully explained by uncontrolled market-beta exposure (long  $\approx +0.8$  beta, short  $\approx -1.2$  to  $-1.4$  beta) in markets that rose  $\sim 20\%$ /year over the sample — once beta is controlled for, **alpha is insignificant in all 12 regressions tested, both legs, both markets, both frequencies**. There is currently no demonstrated stock-selection skill in this signal, in either direction. **Corrected rule:** this signal is not a demonstrated source of alpha as currently built; using it as a behavioral tilt requires beta-neutralizing both legs first and re-testing the hedged residual for alpha — done in Part V.8, which independently confirmed the same conclusion with an actual rolling, out-of-sample hedge rather than a single regression coefficient. And more generally: check for a beta mismatch between a long-short book's legs before reaching for a behavioral explanation, not after three rounds of increasingly exotic ones.

**UPDATE FROM PART V.9**

Part V.9 applied the same beta check to the other two signals, and found something important in both directions. Short-term reversal — this project's one previously-reported positive finding — failed every one of its 12 alpha tests once beta was controlled for. It had never been re-checked with the standard of rigor developed for the losing signal; once it finally was, it didn't hold up either. **Rule: a positive finding earns no exemption from the checks a negative one gets.** But 12-1 momentum, tested the same way, showed something new: on the US mirror, its long leg and combined book carry large, highly significant alpha (annualized roughly +8-15%/yr,  $p < 0.01$  in every cut), with the combined book's beta close to zero. This is currently the project's single most credible candidate for a genuine, demonstrated edge — not replicated on NSE, but a coherent, high-significance cluster rather than the isolated marginal hits this project has learned to distrust.

**UPDATE FROM PART V.10**

Part V.10 checked the one thing Part V.9 left open: publication decay. Real, textbook-consistent decay was found — alpha down 30-47% post-1994, one cut (combined book, daily) losing significance entirely — but not disappearance: three of four cuts remain significant through 2017 alone. **Size any claim about this finding on the decayed, post-1994 number, not the full-sample one.**

**UPDATE FROM PART V.11**

Part V.11 re-tested Part V.10's post-1994 number with an actual rolling, out-of-sample hedge (Part V.8's methodology) instead of an in-sample regression, and found it does not survive: post-1994 alpha is not statistically significant in either leg, at either frequency, once beta is estimated only from trailing data and re-hedged monthly. Pre-1994 alpha remains strongly confirmed. Treat this project's momentum finding as real and robust pre-1994, and not currently demonstrated to be forward-sizeable in the post-publication era.

**UPDATE FROM PART V.12**

Part V.12 dug into that null result at explicit request rather than stopping here: is it really just the 2009 momentum crash, or the hedge's rolling window? Neither explained it — a five-era breakdown showed ongoing decay through 2017, well after 2009, and the result held across three different hedge windows. This is the most rigorous read of momentum specifically: genuine, strong pre-1994 alpha, and genuine, ongoing decay since — not a temporary shock the strategy is due to recover from.

**UPDATE FROM PART V.13**

Part V.13 then asked whether this whole pattern is unique to momentum. It isn't: short-term reversal's long leg shows the identical "real pre-1994, decayed since" signature ( $p=0.046$  pre-1994,  $p=0.629$  post), invisible inside Part V.9's full-sample null. 52-week-high, by contrast, shows no significant alpha in either era — it never had genuine skill at all. **The decay pattern is market-wide, appearing in two of three signals' long legs, not a momentum idiosyncrasy — and Part V.9's reversal retraction should be read as accurate at the full-sample level but incomplete.**

## 6.2 A risk-management playbook, from the Q1 simulation

- **Never calibrate tail risk on a calm-regime correlation matrix alone.** The Part III simulation shows a Gaussian VaR model calibrated on "normal" data understates the true 99.9% tail loss by roughly 1.3x, purely because it cannot see correlations rising toward 1 under stress. Any leveraged strategy needs a stress-regime correlation scenario as a second, explicit check, not just a historical-calibration number.
- **Leverage does not just scale losses — it changes which losses are survivable.** LTCM's underlying spread moves were not physically unprecedented; 25:1+ leverage turned a bad quarter into insolvency. Position and leverage limits should be set against the stress-regime tail estimate, not the calm-regime one.
- **Crowding is a correlation risk invisible from inside your own portfolio.** The 2007 Quant Quake shows a well-diversified-looking book can be secretly correlated with every other fund running a similar signal. A practical mitigant: track how crowded a factor appears to be (e.g., the realized correlation of a strategy's returns to a public factor index) and de-risk when it's high, independent of the strategy's own apparent confidence.
- **A winning streak is not evidence against tail risk.** Amaranth's Brian Hunter had a strong record before the 2006 blowup; recent success under one regime is weak evidence a concentrated position is safe under a different one. Concentration limits need an explicit override that doesn't relax just because a book has been working.
- **Treat "the model says it's fine" as a hypothesis, not a fact** — especially near known regime-change triggers (sovereign defaults, liquidity crunches, crowded-factor unwinds). This is the project's central thesis applied reflexively to its own tooling, not just to the markets it studies.
- **The Q1 mechanism and the Q2 investigation asked the same question — Q1 answered it with a simulation, Q2's real data gave a more honest, weaker answer.** Part III's simulation showed, stylized, that tail risk compounds when volatility and correlation rise together. Part V.5's momentum-crash-risk investigation set out to find the same signature in a real backtest, and a first descriptive pass looked like it had. Formal significance testing (Part V.6) did not confirm it. Any short position built on a behavioral signal should still be regime-tested before being sized — that precaution doesn't depend on this specific mechanism being confirmed — but "regime-tested" has to mean the HAC-regression version, not the regime-bucket version, given what happened here when the two disagreed.

- **A compelling descriptive pattern is a hypothesis, not a finding — this project produced its own cautionary tale.** Part V.5's regime-bucket comparison looked like a clean, four-signature confirmation of momentum-crash risk. Formal HAC-regression testing at daily and monthly frequency (Part V.6) found none of the regime-conditioning coefficients significant in the leg the theory actually predicts, in either market. Both analyses used the same underlying data; only the statistical rigor differed. Never size a position, write a risk limit, or make a claim based on a descriptive regime split alone — run the regression with proper standard errors first, and expect a real chance the compelling-looking pattern won't survive it.
- **The cheapest test is the one to run first, and this project ran it last.** Three escalating hypotheses (crash-window concentration, a value/growth confound, formal momentum-crash regime-conditioning) were tested across three milestones before Part V.7 finally ran a plain CAPM beta regression — the single cheapest, most standard check for any long-short book — and found the entire answer in it: an uncontrolled beta mismatch between the legs, fully explaining every prior milestone's numbers, no behavioral story needed. Order hypothesis tests from cheapest and most mechanical to most exotic, not the reverse; a beta regression takes minutes and rules out the most common cause of surprising long-short performance before any more elaborate explanation is worth reaching for.
- **A risk process that only re-tests its losers eventually trusts a winner it never should have.** Every rigor upgrade through Part V.8 was applied to the signal that was losing money. The one signal reported as a genuine edge (reversal) rode on its original, less rigorous validation for six milestones before anyone checked it the same way. Part V.9 finally did, and it didn't hold up. Schedule periodic re-validation of every "confirmed" edge under your *current* standard of rigor, not just the standard at the time it was confirmed — a standard that improves over a book's life should apply retroactively, especially to positions still being sized on the old conclusion.
- **Size a position on the decayed number, not the full-sample number — and check that decayed number was itself estimated out-of-sample.** Part V.10 split the one surviving edge (US momentum) at its 1993 publication date and found real decay: alpha down 30-47% post-1994, one cut losing significance outright. That was already a large correction — but it was still an in-sample estimate. Part V.11 re-tested the same period with an actual rolling, out-of-sample hedge and found the post-1994 alpha does not survive at all, in either leg. The full-sample average was never the honest number to size against once a signal is old enough to have a publication date, and an in-sample per-era regression isn't the final word either — only a hedge that could actually have been traded forward tells you what capital entering today can expect.
- **Before accepting a null result, rule out the obvious "maybe it's just one bad episode" and "maybe it's a methodology artifact" objections — and if it survives both, treat that as the null result getting stronger, not weaker.** Part V.12 tested whether Part V.11's post-1994 null result was really just the 2009 momentum crash, and whether it depended on the specific hedge window chosen. Neither objection explained it: the era-by-era trend showed ongoing decay through 2017, well after 2009, and the result was identical across three different hedge windows. A finding that survives a genuine attempt to explain it away deserves more confidence, not less — the temptation after an unwelcome result is to look for the one adjustment that makes it go away; running that check honestly, and reporting it even when it doesn't help, is what separates a stress-test from a fishing expedition.

- **A full-sample "no significant alpha" verdict does not mean a signal was never real — check whether it's actually two eras averaging to zero before writing it off entirely.** Part V.9 declared short-term reversal fully retracted based on a full-sample regression, the technically correct read of that specific test. Part V.13 applied the era-split toolkit built for momentum and found reversal's long leg had been genuinely, significantly positive pre-1994 and decayed to noise since — the same pattern as momentum, hidden inside a full-sample average that happened to net out near zero. Retest every full-sample null result from before your era-split toolkit existed with that toolkit, not just your full-sample findings — a "no effect on average" verdict can quietly contain a real, decayed effect that a single-number summary cannot distinguish from a signal that was simply never real.

## 6.3 A standalone business idea: decomposed behavioral signal analytics

Off-the-shelf factor data (momentum, value, quality) is typically sold by large vendors as pre-blended, black-box composites, priced for institutions with large budgets. Smaller systematic funds, family offices, independent RIAs, and research desks either cannot afford that tier, or cannot see *inside* the composite to know which component is actually carrying the edge on their specific universe — exactly the failure mode this project hit directly on the NSE data (Part V.3), and hit again when its own first-reported positive finding (reversal) failed a beta check the project hadn't yet thought to run (Part V.9). Decomposed, beta-adjusted validation is what surfaced the project's one genuine finding (US momentum) instead of its retracted one.

### DECISION & RATIONALE

This idea was originally framed as an extension of the specific software platform this research happened to be built alongside. That framing was dropped after explicit feedback that the business idea should stand on its own, independent of any particular existing product. The pitch below reflects that: a standalone service, differentiated on the same transparency principle (decomposed, auditable signals; never a pre-blended black box) that Part V.3's finding demonstrated the value of directly.

**The product:** a subscription analytics service with two parts — a **signal side** (the momentum / 52-week-high / reversal library, run per client against *their* universe, reported as separate, individually-backtested components, never pre-blended, each with its own beta-regression alpha/beta breakdown so a client can see whether performance is genuine stock-selection skill or just uncontrolled market exposure — Part V.7's finding made this check a required feature, not an optional one — and, for any signal drawn from published academic research, a pre/post-publication decay split re-confirmed with an actual out-of-sample hedge, not just an in-sample regression (Parts V.10-11's check), a second differentiator most factor-data vendors don't surface at all) and a **risk side** (the regime-switching stress-VaR methodology from Part III, run against a client's actual position correlations and leverage, reporting calm-regime vs. stress-regime tail loss side by side).

**Target customer:** small-to-mid systematic equity funds, family offices, and independent RIAs priced out of institutional factor-data tiers but sophisticated enough to want decomposed, re-validated signals rather than a black box; secondarily, finance graduate programs and CFA/PE prep courses, as a teaching tool for the exact "don't trust the blend" lesson this project surfaced.

**Revenue model:** a per-seat analytics subscription, tiered by number of tracked universes/portfolios — the same go-to-market as existing quant factor-data vendors, priced and scoped for the segment those vendors serve poorly, differentiated specifically on transparency: every number traces to runnable code and a stated backtest window, not a proprietary methodology.

**Moat:** not the signals themselves (all public, published research) — the moat is the discipline of per-client, per-universe decomposed validation instead of a generic pre-blended score, which is exactly what a larger vendor selling a standardized product across all clients structurally cannot do cheaply.

## Part VII — Limitations and Honesty Statement

Stated together in one place because a project whose entire thesis is "don't trust a model's output just because it looks rigorous" owes the same discipline to its own outputs.

- **Survivorship bias.** Both the NSE mirror and the US mirror located in milestone 3 use a present-day-chosen company list, not point-in-time index membership. A rigorous version needs a survivorship-bias-free universe (e.g., CRSP, or a maintained point-in-time constituents file).
- **Data provenance.** Both the NSE and US mirrors are personal, community-uploaded GitHub repositories, not official exchange or licensed vendor feeds. Results from them are a genuine methodology demonstration on real prices, not investment-grade research, until cross-checked against an official source.
- **Momentum and reversal did not replicate consistently** across the two markets tested (Part V.4) — treat either result, in isolation, as provisional rather than a confirmed anomaly. Only the 52-week-high signal's loss held up in both.
- **Short-term reversal's positive result is retracted at the full-sample level (Part V.9) — but its long leg specifically is a decayed, not a never-real, effect (Part V.13).** The same CAPM beta check that debunked the 52-week-high signal, applied to reversal for the first time, found none of its 12 full-sample alpha tests significant. That verdict is accurate but incomplete: Part V.13 later found reversal's long leg carries real, significant pre-1994 alpha ( $p=0.046$ ) that decays fully to noise post-1994 ( $p=0.629$ ) — the identical pattern found for momentum, invisible in a single full-sample average. Its short leg and combined book remain non-significant in every era tested.
- **US 12-1 momentum's post-1994 alpha does not survive an actual out-of-sample hedge, and the deeper investigation confirms genuine decay rather than a crash or methodology artifact (Parts V.10-12, superseding an earlier, more optimistic in-sample reading).** An in-sample pre/post-1994 split (Part V.10) found real decay but apparent survival in three of four cuts. Re-testing the same post-1994 period with a rolling, out-of-sample beta hedge found no statistically significant alpha in either leg, at either frequency (Part V.11); the hedged combined book's post-1994 average return is outright negative. Part V.12 then tested whether that null result was just the 2009 momentum crash or a hedge-window artifact: an era-by-era breakdown showed ongoing decay through 2017 (well past 2009), excluding the crash window left the combined book's result essentially unchanged, and the null result held across three different hedge windows. Pre-1994 alpha remains real and robust under every test. This project's momentum finding should be read as strong and genuine before 1994, and as genuine, ongoing decay — not a temporary shock — since.
- **"Real pre-1994, decayed since" is not momentum-specific (Part V.13).** Applying the identical pre/post-1994 out-of-sample hedge to all three US signals found two distinct stories: 52-week-high shows no significant alpha in either era, in any leg (it never had genuine stock-selection skill); reversal's long leg shows the identical decay signature as momentum's. The pattern appears in two of three signals' long legs, not one, consistent with a market-wide explanation rather than a momentum idiosyncrasy.
- **The 52-week-high result's cause: resolved, and it's the boring answer.** Two behavioral/statistical explanations were ruled out by direct test: crash-window concentration (Part V.4)

and a value/growth confound (Part V.5), neither surviving in either market; formally-tested momentum-crash risk (Part V.6) then walked back Part V.5's descriptive-looking confirmation too. The actual explanation (Part V.7) is a construction flaw, not a market phenomenon: the legs have significantly different, uncontrolled market-beta exposure (long  $\approx +0.8$ , short  $\approx -1.2$  to  $-1.4$ ), leaving the book net short-beta ( $-0.32$  to  $-0.70$ ,  $p < 0.001$  every cut) in markets that returned  $\sim 20\%$ /year. Once beta is controlled for, alpha is insignificant in all 12 regressions tested. There is no demonstrated stock-selection skill in this signal, in either direction, as currently built.

- **Transaction costs are a simple linear model**, not a real market-impact model; a strategy sized for real capital would need a proper implementation-shortfall estimate.
- **No out-of-sample / walk-forward validation is wired up by default.** A real deployment should split into a strict in-sample fit period and out-of-sample test period, and check for performance decay after each anomaly's academic publication date — a well-documented risk for momentum and reversal specifically.
- **The Q1 simulation's parameters are illustrative**, calibrated loosely to the LTCM episode's qualitative shape (correlations that were low/moderate in normal times moving toward 1 in crisis), not fit to LTCM's actual, never fully disclosed book. Its reported multiple should be read as an order of magnitude, not a precise historical reconstruction.
- **The beta-hedge test (Part V.8) is itself imperfect, by design and by honest admission.** The rolling hedge leaves a small residual correlation with the market ( $-0.02$  to  $-0.07$ , not exactly zero), the estimated beta is quite unstable across time in both markets (its standard deviation rivals its mean), and no hedging transaction costs are modeled. None of that changes the direction of the finding (still no significant residual alpha), but a live implementation would need a more robust hedging scheme and would bear real costs this analysis doesn't capture.

# Conclusions

The Efficient Market Hypothesis is a useful default, not a law of nature. This project's five case studies show it failing in genuinely different ways — a Nobel-laureate-staffed fund undone by other investors' correlated panic; sophisticated quant funds undone by their own hidden crowding; an entire industry's risk model undone by a correlation assumption nobody's calibration data ever tested; a struggling retailer's stock undone (in the institutional short-sellers' favor, this time) by coordinated retail attention — which is itself evidence against any single, tidy story of "why markets are irrational." There isn't one mechanism; there are several, each traceable to a specific, well-documented piece of human psychology.

The project's real empirical result (Part V.3) delivered a finding more useful than a simple confirmation of either extreme view would have been: not "markets are irrational, biases always pay," and not "the anomalies are all arbitrated away, don't bother." Instead: one specific, named bias (anchoring, via the 52-week-high signal) actively lost money on this universe, while a different one (overreaction, via short-term reversal) at first looked like a real, cost-adjusted edge — and the two were invisible from inside a single blended composite. That is, in miniature, the entire project's argument: rigor means checking the specific mechanism, not trusting the reassuring-looking aggregate. It took until Part V.9 for the project to apply that same rigor to its own positive-looking half of the story, and when it did, the "edge" didn't survive either — but a different signal, US 12-1 momentum, did, and Part V.10 went one step further than any earlier milestone: instead of asking whether the finding was real, it asked how much of it a trader could still expect to capture today, and found a genuinely mixed, honest answer at first — real decay, not disappearance. Part V.11 pushed the same question one step further still, replacing an in-sample estimate with an actual out-of-sample hedge, and the honest answer got more sobering: no statistically significant post-1994 alpha survives in either leg. Even this project's own most reassuring recent finding kept shrinking every time it was checked with a sharper tool. Part V.12 then did something different from every check before it: instead of applying a sharper tool and finding the result shrink again, it tried directly to explain the shrinkage away — as a single crash episode, or as an artifact of one modeling choice — and failed on both counts. The result held. Part V.13 then went back to the signal Part V.9 had seemed to retract outright — short-term reversal — and applied the same era-split toolkit rather than trusting the earlier full-sample verdict, and found reversal's long leg had been telling the truth about being "a real, cost-adjusted edge" all along, just not for the era that mattered by the time anyone checked: genuine pre-1994, decayed since, the same shape as momentum's story.

The practical output (Part VI) turns that into three concrete artifacts: an investment framework that explicitly forbids trusting a blend without decomposing it; a risk-management playbook built directly from a real simulated result, not a generic checklist; and a business idea whose differentiation *is* the decomposition discipline the research itself needed. Milestones 3 through 12 then ran the India findings through increasingly rigorous versions of the same skepticism the project applies to everything else, and at every step a stronger method found something the weaker one had missed or overclaimed: momentum and reversal turned out to be universe-dependent, not general truths; a value/growth confound was tested and rejected; a formally-compelling-looking momentum-crash mechanism was tested and rejected too, once proper statistical significance testing replaced descriptive pattern-matching; the actual explanation for the losing signal, when it finally arrived, was the most basic and least glamorous of all the candidates tested — an uncontrolled beta mismatch, checked last instead of first, then confirmed a second way with

an actual hedged strategy; and then, applying that same beta check to the signal the project had been treating as a validated success, the success turned out not to be one — while a signal nobody had flagged as special (US momentum) turned out to hold real, statistically robust alpha once someone finally looked properly — alpha that Part V.10 then showed had partially decayed since its 1993 publication, that Part V.11 then showed does not clear this project's own bar for a forward-sizeable edge once tested with an actual out-of-sample hedge, that Part V.12 then confirmed is genuine, ongoing decay rather than an artifact of the 2009 crash or the hedge's own design, and that Part V.13 then showed is not even unique to momentum — the identical shape turned up in a signal this project had already, separately, declared dead.

The fix that survived all of that scrutiny is more modest, and the project's real positive finding is broader and more precisely dated than any earlier draft of this conclusion claimed: there is no demonstrated, beta-independent stock-selection skill in the 52-week-high signal, in either direction, as currently built, at any point in the sample. Momentum's long leg and short-term reversal's long leg, by contrast, both show robust, largely beta-independent edges — on the US mirror specifically, and only demonstrated to hold in the era before roughly the mid-1990s. Every one of those findings was reached not by anyone's original hypothesis about which signal should work, but by finally applying the project's own hard-won standard of rigor evenly: to a winner as well as a loser, to a signal's own more recent, more comfortable-looking result as well as its oldest one, and — last of all — to a signal already written off, instead of assuming a full-sample null closed the question for good. That is the project working as intended, including on itself: not every finding needs to be confirmed to be useful, a compelling pattern — positive or negative — is a hypothesis until it survives testing at every level of rigor available, and the single most important thread running through this entire guide is a project that kept correcting its own most recent, best-supported-looking result, seven times in a row — six outright retractions or downward revisions, and one nuanced check (Part V.10) that briefly looked like a stopping point before Part V.11 showed it wasn't — then ran an eighth check (Part V.12) built specifically to try to explain the seventh correction away, and couldn't, and a ninth (Part V.13) that went looking for the same pattern somewhere the project had stopped looking, and found it. Not finding an escape hatch, and finding the same shape twice in independent places, are both findings: the project never found a result durable enough to stop re-checking, and has now shown its central pattern — genuine edge before the mid-1990s, genuine decay since — is more general than any single milestone first suggested, leaving two narrow, well-tested claims (pre-1994 momentum and pre-1994 reversal, both long-leg-only) to currently trust, and one honest, thoroughly-stress-tested absence (the post-publication era, across signals) to size nothing against.

# Glossary

## Alpha

The part of a strategy's average return left over after accounting for its market exposure (beta) -- the piece that would represent genuine stock-selection skill, if it's significantly different from zero. This guide's Part V.7 finds none in the 52-week-high signal, either leg, once beta is controlled for.

## Anchoring

Relying too heavily on an initial reference point (e.g. a 52-week high, an opening price) when making subsequent judgments.

## Beta

How much a position tends to move for a given move in the overall market -- a beta of 1 means it moves with the market on average, above 1 means it amplifies market moves, negative means it moves opposite the market. A long-short book with unequal long and short beta carries unintended net market exposure.

## Composite score

A single number blending several individual signals into one rank; this project's central finding is that a composite can hide a component that is actively harmful.

## Correlation regime shift

A change in how much different assets move together — typically a jump from low/moderate to near-1 correlation during a market crisis.

## Crowding

Many independent, sophisticated participants arriving at similar positions without coordinating, producing hidden correlation across supposedly diversified portfolios.

## Decile / quintile backtest

A methodology that ranks a universe by a signal, then goes long the best-ranked group and short the worst-ranked group, to isolate the return spread the signal actually predicts.

## Efficient Market Hypothesis (EMH)

The theory that asset prices reflect all available information, implying no strategy should reliably beat a passive index.

## Extrapolation bias

Assuming a recent trend (e.g. fast earnings growth) will continue further into the future than it typically does.

## Fat tails

A return distribution with more extreme outcomes than a normal (bell-curve) distribution predicts.

## Herding

Following the crowd's behavior rather than independently evaluating a decision — useful most of the time, dangerous when the crowd itself started moving for an unrelated reason.

## Leverage

Borrowed capital or notional derivative exposure used to scale a position beyond what the investor's own capital would allow; it magnifies both gains and losses.

**Maximum drawdown**

The largest peak-to-trough decline a strategy experiences at any point in a backtest.

**Momentum**

The tendency of a security's recent trend to continue, typically attributed to underreaction.

**Momentum crash risk**

The tendency of a "long recent winners, short recent losers" strategy to suffer severe losses when the short leg's high-beta losers rebound sharply, typically during high-volatility, post-downturn market recoveries.

**Overconfidence**

Overestimating the reliability of one's own judgment or model, often reinforced by a recent string of favorable outcomes.

**Post-earnings-announcement drift (PEAD)**

The tendency of a stock price to keep drifting in the direction of an earnings surprise for weeks after the announcement, rather than repricing immediately.

**Newey-West (HAC) standard errors**

A correction to a regression's standard errors that accounts for serial correlation in the data (e.g. from overlapping monthly holding periods); without it, significance tests on this kind of strategy-return data can look more confident than the data actually supports.

**p-value**

The probability of seeing a result at least this extreme if there were truly no effect. Conventionally,  $p < 0.05$  is called "statistically significant" -- but with many tests run at once, some will cross that bar by chance alone (see this guide's own Milestone 5 for a worked example).

**Reversal**

The tendency of a very recent, sharp price move to partially revert, typically attributed to overreaction.

**Sharpe ratio**

Average return divided by return volatility, annualized; a standard measure of risk-adjusted performance.

**Survivorship bias**

The distortion introduced by only including, in a historical study, entities that are still around today (e.g. still-listed companies), which tends to overstate historical performance.

**Value-at-Risk (VaR)**

An estimate of how much a portfolio could lose over a given horizon at a given confidence level (e.g. the worst-case loss on all but the worst 1% of days).

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*All code, full case-study write-ups, and the underlying data-loading and backtest engine referenced throughout this guide live in the project's code repository, folder research/behavioral-finance/, which remains the definitive, most up-to-date source as milestone 3 and beyond continue.*